FastQC report modules

Module	Description	Quality control status	Troubleshooting
Basic statistics	Simple composition statistics.	"Pass": will always pass. "Warning": never raises a warning. "Failure": never raises an error.	Never raises warnings or errors, therefore no troubleshooting needed.
Per-base sequence quality	Overview of the range of quality values across all bases at each position. Good calls - green (PHRED score >28), moderate calls - orange (PHRED score >20 and ≤28) and poor calls - red (PHRED score <20).	"Pass": lower quartile for any base is ≥10, or if the median for any base ≥25. "Warning": lower quartile for any base is <10, or if the median for any base is <25. "Failure": lower quartile for any base is <5, or if the median for any base is <20.	Trimming at the beginning and end of the reads where the quality begins to fall below PHRED score 28. It is common for quality to drop at the 3' end. "LEADING", "TRAILING" or "SLIDINGWINDOW" could be used.
Per-sequence quality scores	Indicated if a subset of sequences had universally low-quality values.	"Pass": most frequently observed mean quality was ≥27. "Warning": most frequently observed mean quality was <27 (0.2% error rate). "Failure": most frequently observed mean quality was <20 (1% error rate).	If bimodal or complex distribution is seen, quality trimming is needed. "SLIDINGWINDOW" or, as a secondary failsafe, "MINLEN" could be used. "MINLEN" could be used after initial trimming if sequence quality is still low: proceed with caution.
Per-base sequence content	Proportion of each base position for which each of the four normal DNA bases had been called. RNA-seq libraries commonly produce biased sequence composition, especially at the start of the read.	"Pass": difference between A and T, or G and C is ≤10% in any position. "Warning": difference between A and T, or G and C is >10% in any position. "Failure": difference between A and T, or G and C is >20% in any position.	Trimming is not recommended since it cannot correct the small sequence bias seen at 5' end of reads. This bias is particularly common in RNA-seq libraries. "HEADCROP" may be used in severe cases, where the bias extends far into the read, but proceed with caution.
Per-sequence GC content	Measured the GC content across the entire length of each sequence and compared it to a modelled normal distribution of GC content. A normal library would produce a roughly normal distribution of GC content.	"Pass": sum of the deviations from normal distribution represents ≤15% of the reads. "Warning": sum of the deviations from normal distribution represents >15% of the reads. "Failure": sum of the deviations from normal distribution represents >30% of the reads.	Whilst a shifted or bimodal curve is a common issue, trimming is not recommended. However, further investigation of potential contamination should be investigated, particularly in the overrepresented sequences module, and can be BLASTed to identify the organism or RNA species.
Per-base N content	Percentage of base calls at each position for which a N was called. If a sequencer cannot make a base call with sufficient confidence, then it will typically substitute the conventional base call with a N.	"Pass": any position shows an N content ≤5%. "Warning": any position shows an N content >5%. "Failure": any position shows an N content >20%.	Low quality bases that increase the frequency of ambiguous bases (N) should be trimmed. "SLIDINGWINDOW", "LEADING" and "TRAILING" could be used. "MINLEN" can be applied if, after initial trimming, the reads have become too short.
Sequence length distribution	Distribution of fragment sizes in the file which was analysed.	"Pass": all the sequences have the same length. "Warning": sequences not all the same length. "Failure": any of the sequences have zero length.	Raw reads should all be one length since they have just been sequenced. After trimming, reads could be multiple lengths. Additional trimming, like "MINLEN" could be used.
Sequence duplication levels	Degree of duplication for every sequence in a library, showing the relative number of sequences with different degrees of duplication.	"Pass": non-unique sequences make up ≤20% of total. "Warning": non-unique sequences make up >20% of total. "Failure": non-unique sequences make up >50% of total.	RNA-seq data typically has some level of sequence duplication, normally from technical duplicates. Therefore, trimming is not recommended as duplicate removal affects downstream analysis. High levels indicate a kind of enrichment bias, like PCR over-amplification.
Overrepresented sequences	All sequences that form >0.1% of the total. For each sequence, the best hit from the database was reported.	"Pass": any sequence forms ≤0.1% of the total. "Warning": any sequence forms >0.1% of the total. "Failure": any sequence forms >1% of the total.	A warning/error may be caused by adapters, primers or rRNA contamination (check with adapter content module). "ILLUMINACLIP", or alignment to a rRNA reference genome and removing them may be used.
K-mer content	Number of each 7-mer at each position in the library. It reported K-mers with positionally-biased enrichment. Distribution of the top 6 most biased K-mers was shown.	"Pass": any K-mer is imbalanced with binomial p ≥0.01 "Warning": any K-mer is imbalanced with binomial p <0.01 "Failure": any K-mer is imbalanced with a binomial p $<10^{-5}$	Warnings or errors may be triggered by adapters, primers or Poly(A/T) tails. Adapter presence should be confirmed with the adapter content module and "ILLUMINACLIP" could be used. If Poly(A/T) tails, no trimming is required.
Adapter content	Identified if there a was significant amount of adapter in a library, thus determining the degree of adapter trimming needed.	"Pass": any sequence is present in ≤5% of all reads. "Warning": any sequence is present in >5% of all reads. "Failure": any sequence is present in >10% of all reads.	Adapters could be trimmed using "ILLUMINACLIP", with optional tightening using "MINLEN" and "TRAILING". Using "TRAILING" can remove low quality bases and "MINLEN" ensures that reads are long enough for alignment.